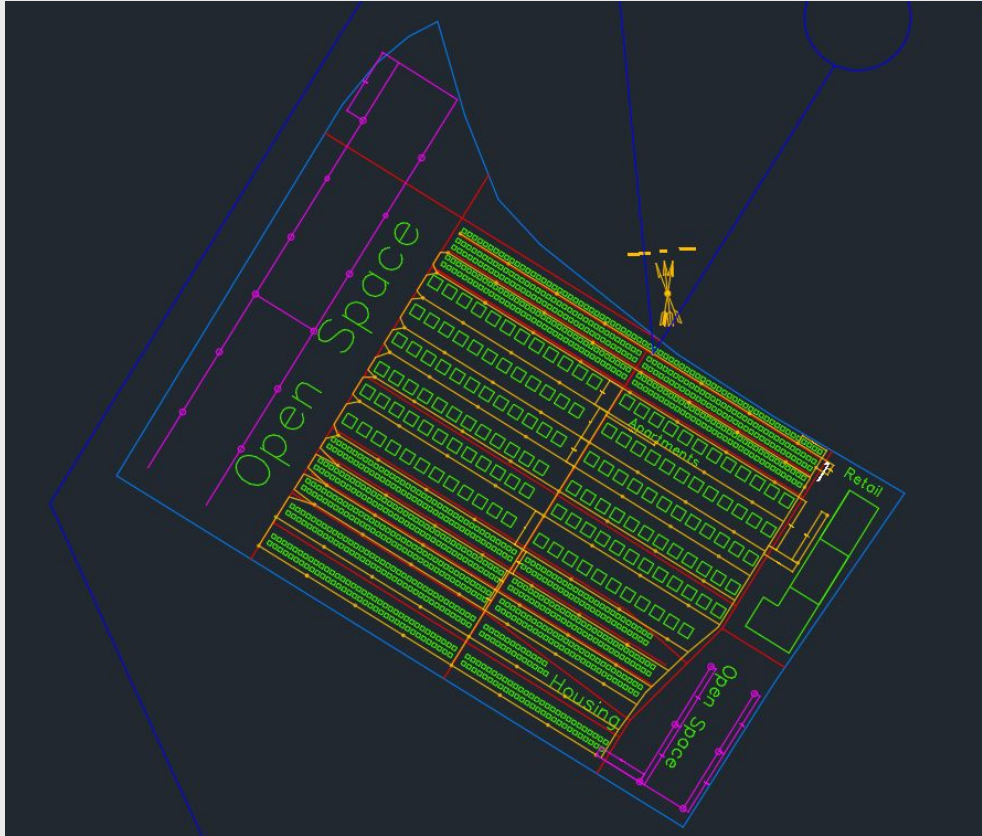


CEE 171 Design Presentation

Amogh Choula, Nathaniel Jordan, Andrew Covarrubias, Matthew Fitzpatrick

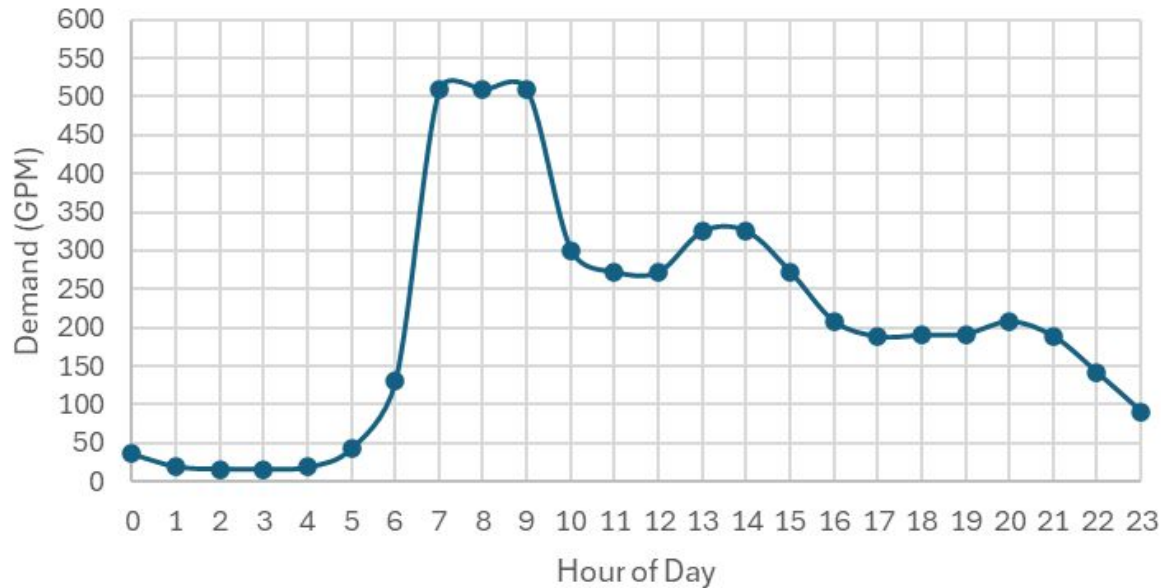
Site Layout



- The given project site is located in Irvine and is roughly 20 million square feet.
 - 8 million square feet is dedicated to open space
 - 1.5 million square feet is dedicated to retail
 - 921 housing units/apartments
 - 1,200 families
- The elevation difference between the southwest boundary of the site and the northeast boundary is 70ft
- Designed to maximize space for households while abiding by design restrictions

Supply and Demand of System (Potable)

Total Residential Demand vs Hour of Day



Details

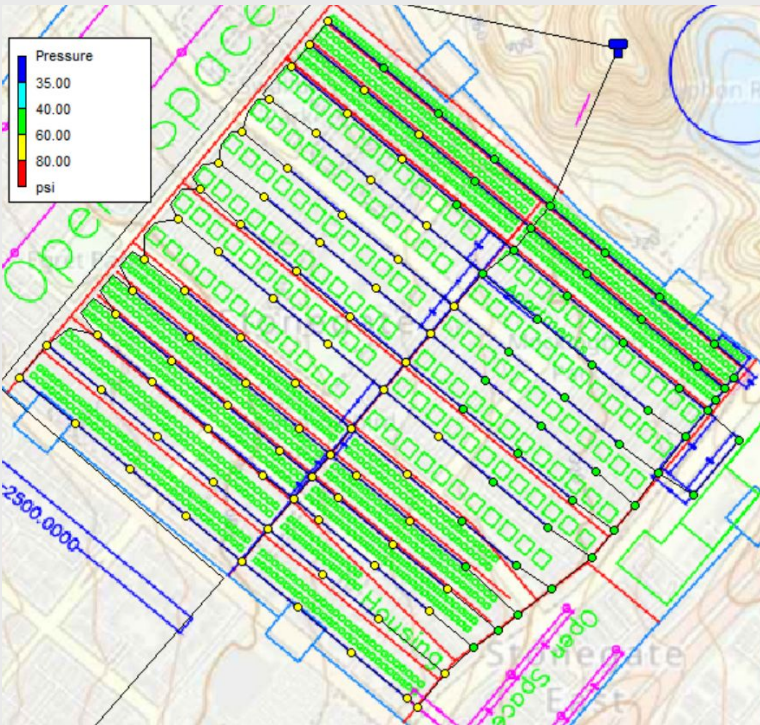
- Total demand for whole community is a little less than 5,000 gallons per day.
- 18 families per node, peak demand from 7am-10am at 4.1 gallons per minute.
- 122 residential nodes and 2 retail nodes throughout the site.

Potable System Constraints

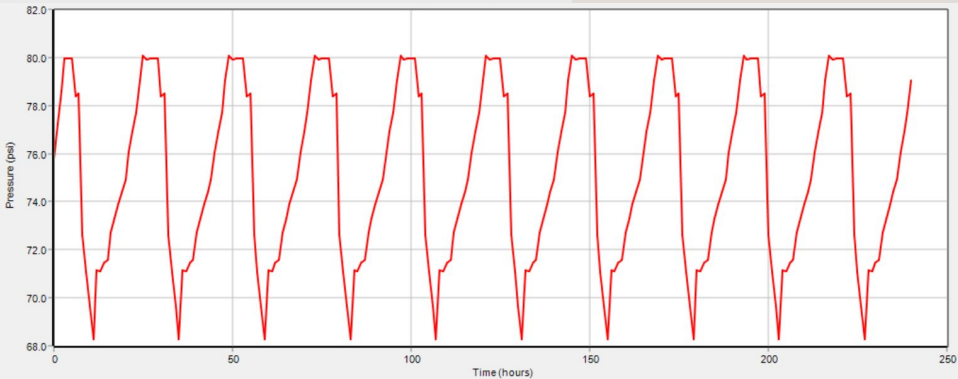


- Pressure: 40psi-80psi
- Well Outflow: $Q_{ave} \leq 250$ GPM
- Residence Time: Age ≤ 48 hr
- Relatively steady tank level
- System demand to be met

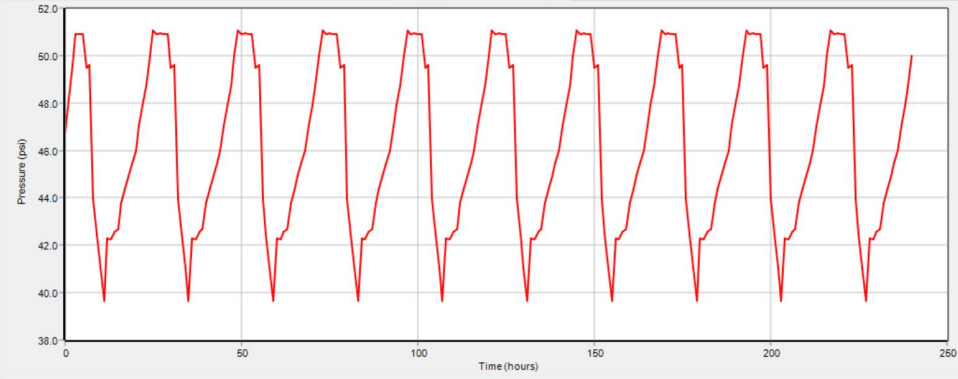
System Pressures



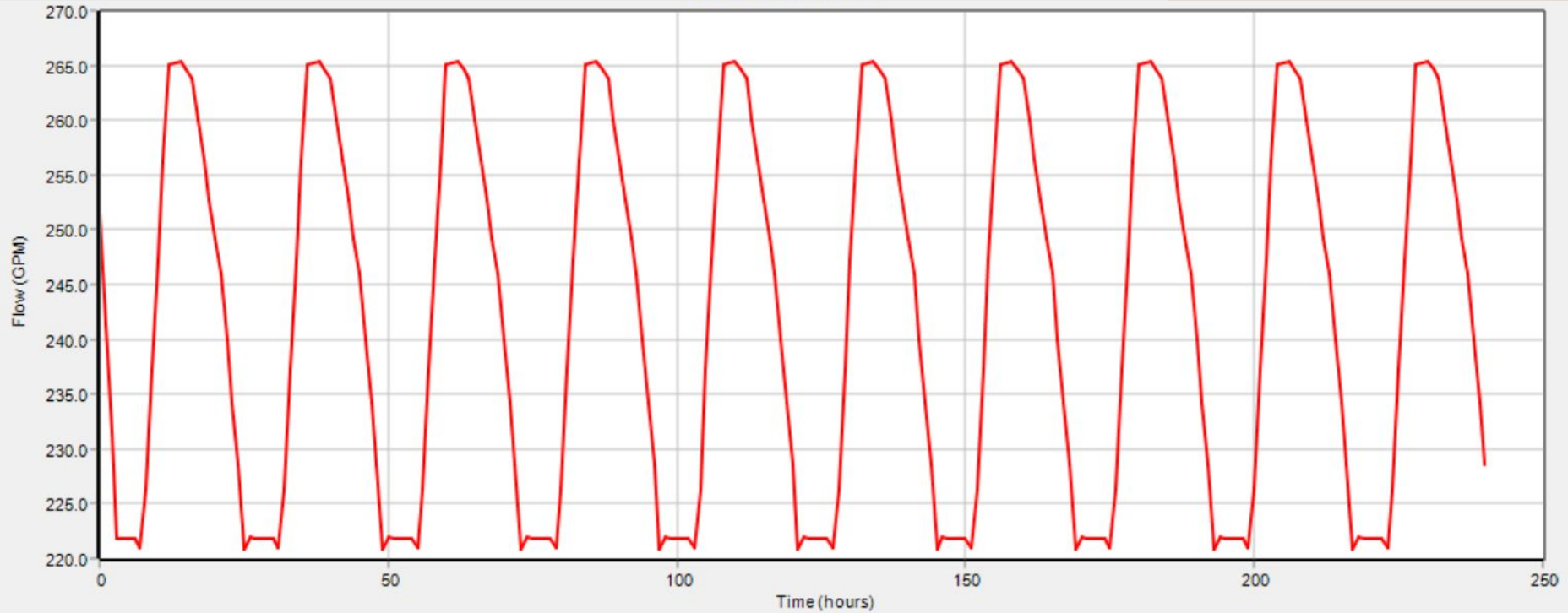
Hourly Pressure (For Highest Elevation Potable Node, 10 days)



Hourly Pressure (For Lowest Elevation Potable Node, 10 days)

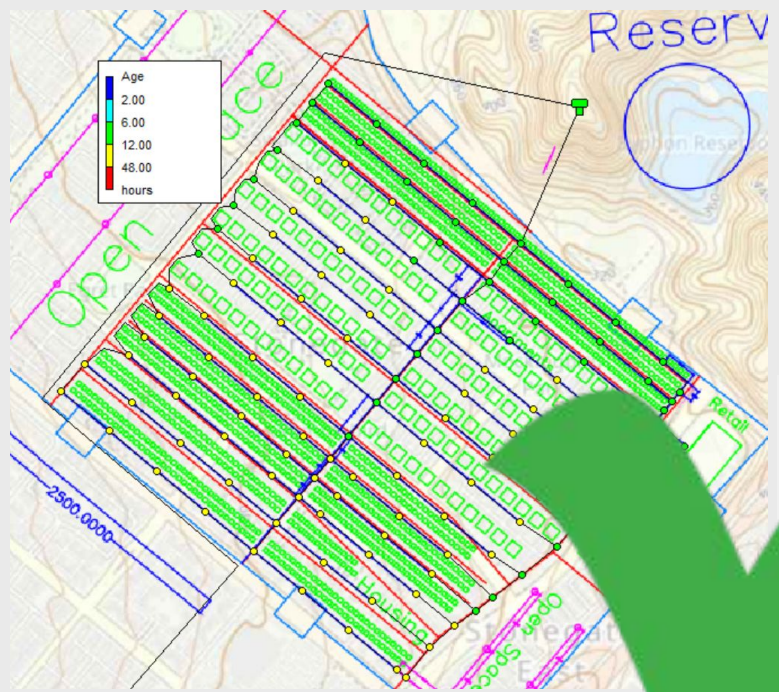


Outflow Of Well (10 Days)

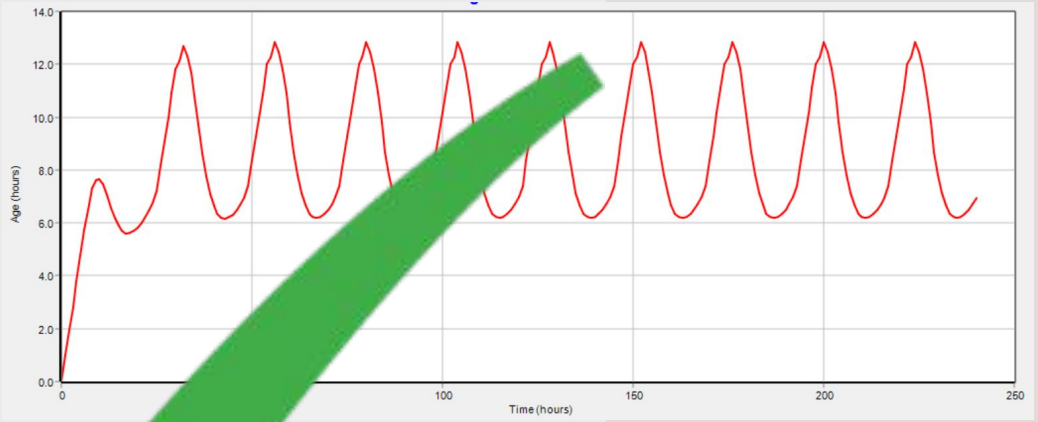


Average Flow = 245 GPM

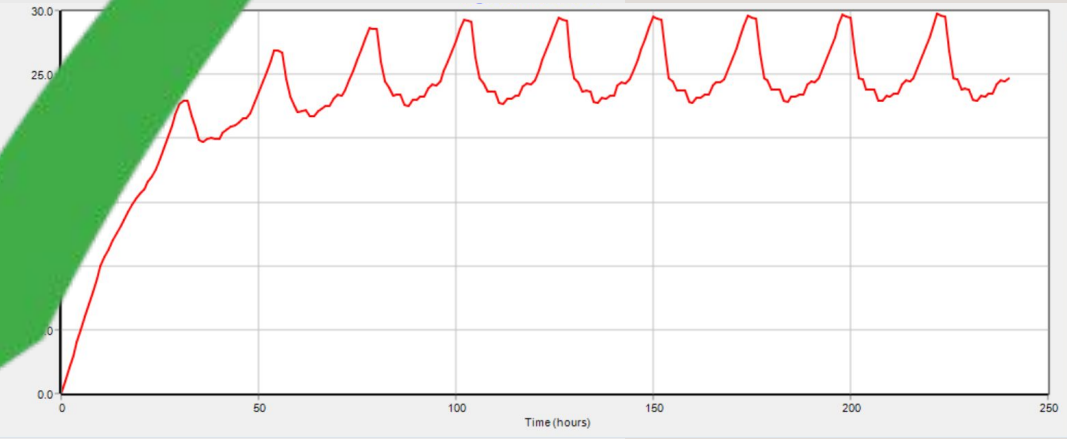
Residence Time



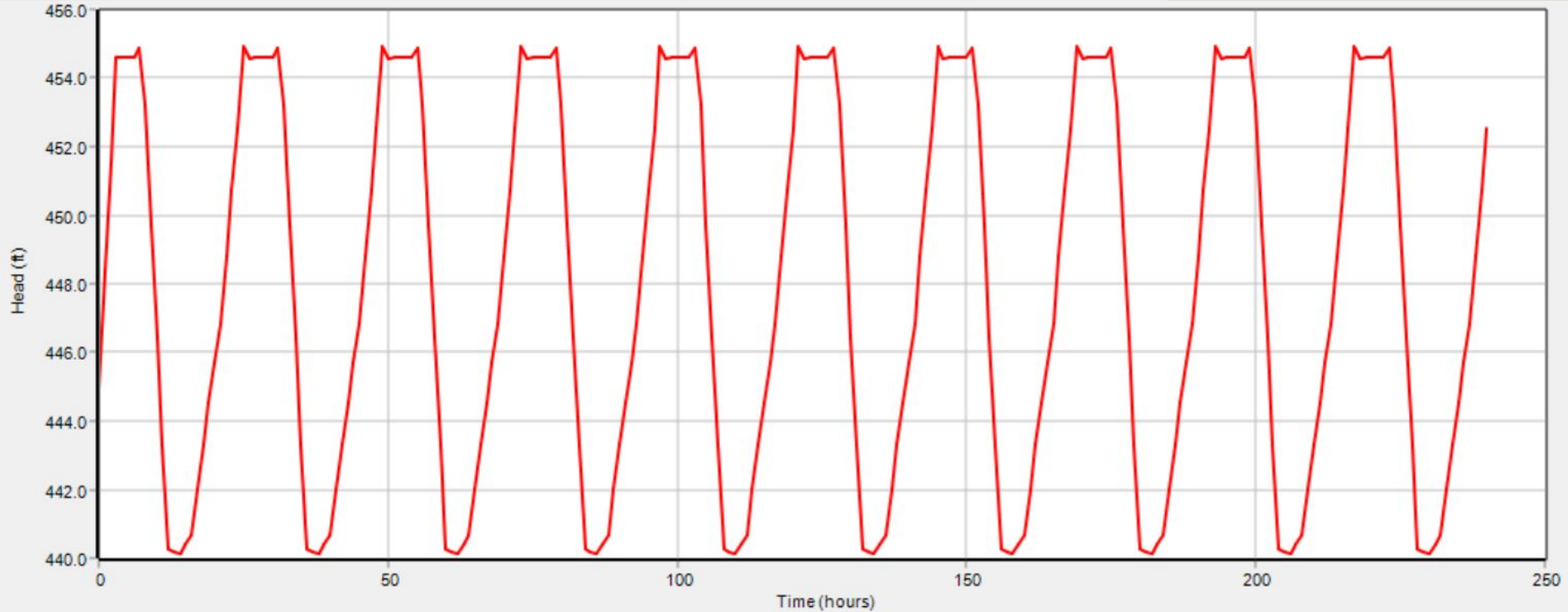
Residence Time Of Potable Tank (10 days)



Residence Time Of Oldest Water Node(10 days)



Hourly Potable Tank Level (10 Days)



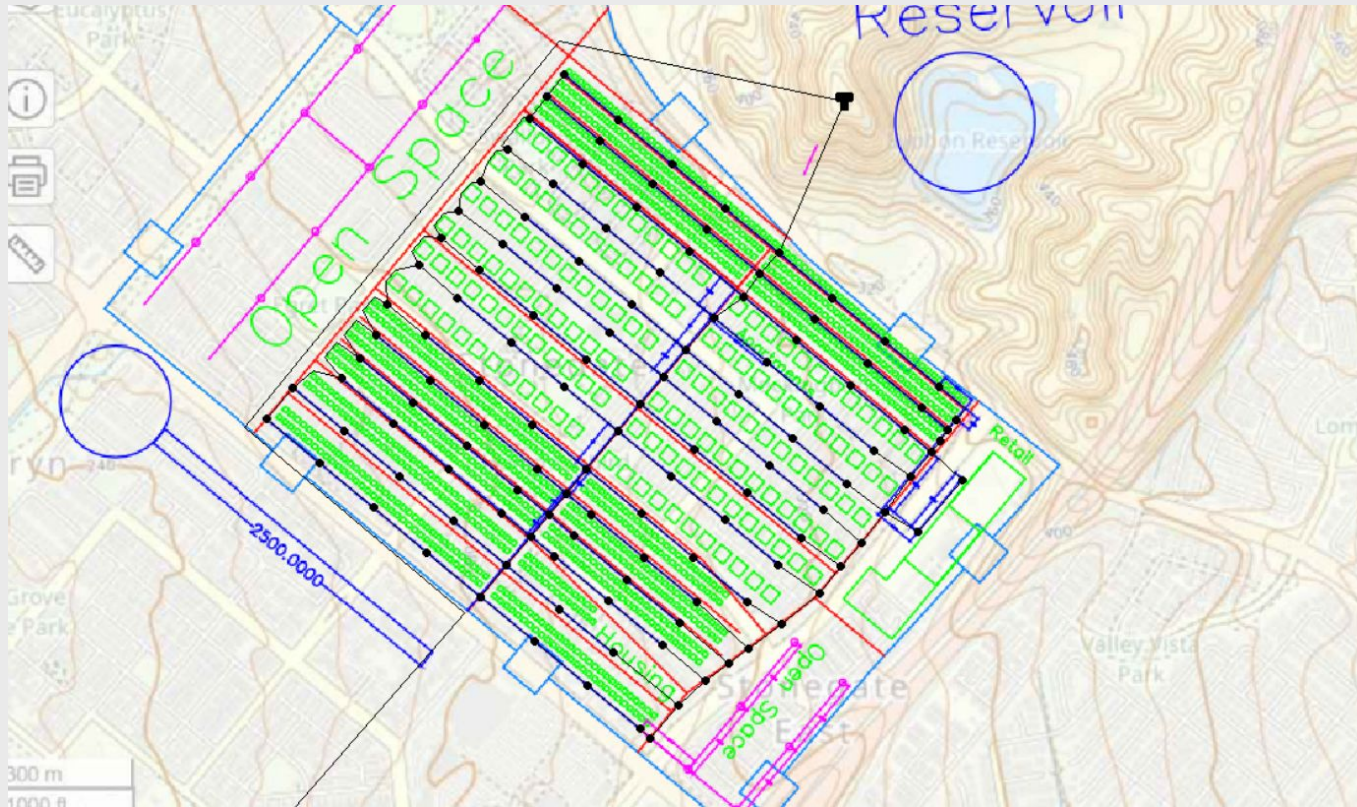
Average Level = 447.5 Feet

Potable Hourly Demand Pattern For One Node (10 Days)



Average Hourly Demand = 1.75 GPM

Design of the Potable Network



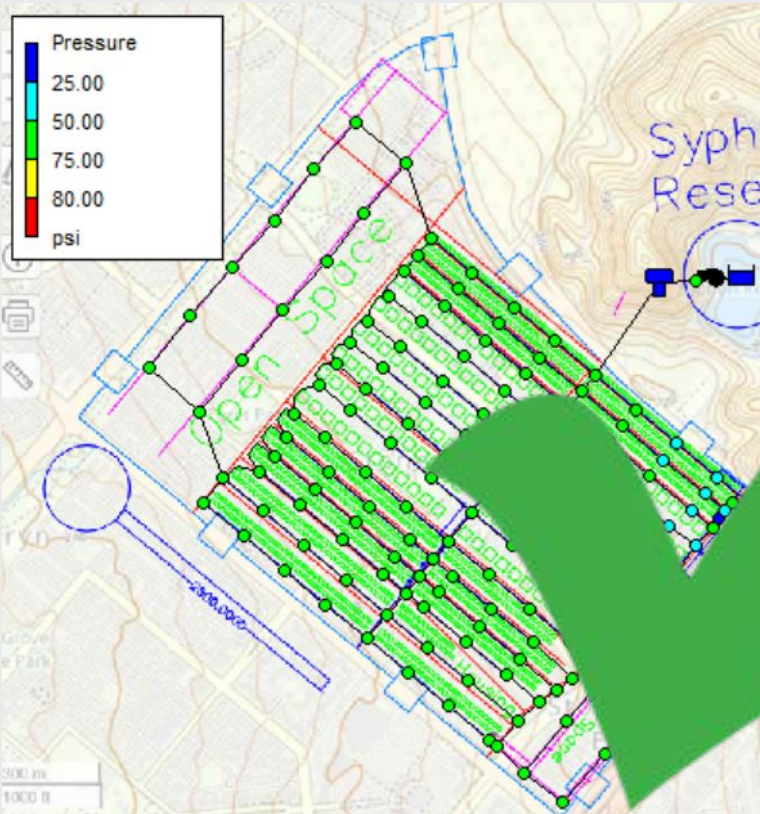
- Gridlike to ensure no dead ends, consistent pressure
- No pipes running under houses
- Storage tank to store water for peak hours
- Some diagonals to reduce pipe length
- Extra "no-demand" nodes for pipe connections

Non-Potable System Constraints

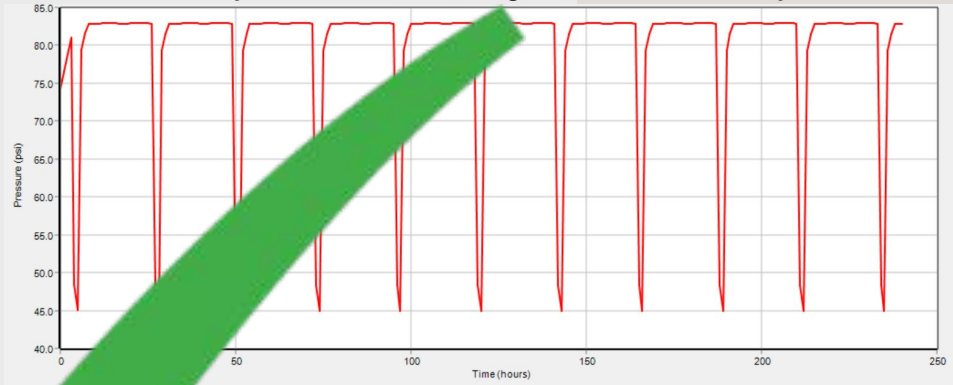


- Pressure must range from 40 - 100 psi
- Must be able to supply a constant 2000 GPM for 2 hours at 2 random residential nodes for fire hydrant
 - Must be during peak season

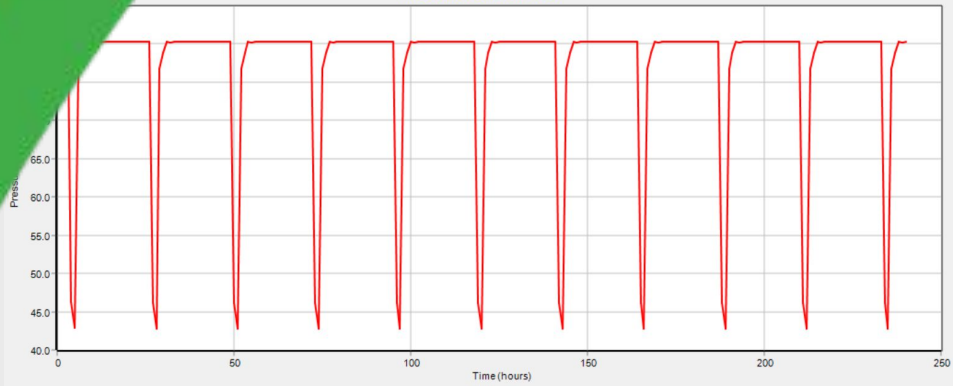
System Pressures



Hourly Pressure (For Irrigation Node, 10 days)

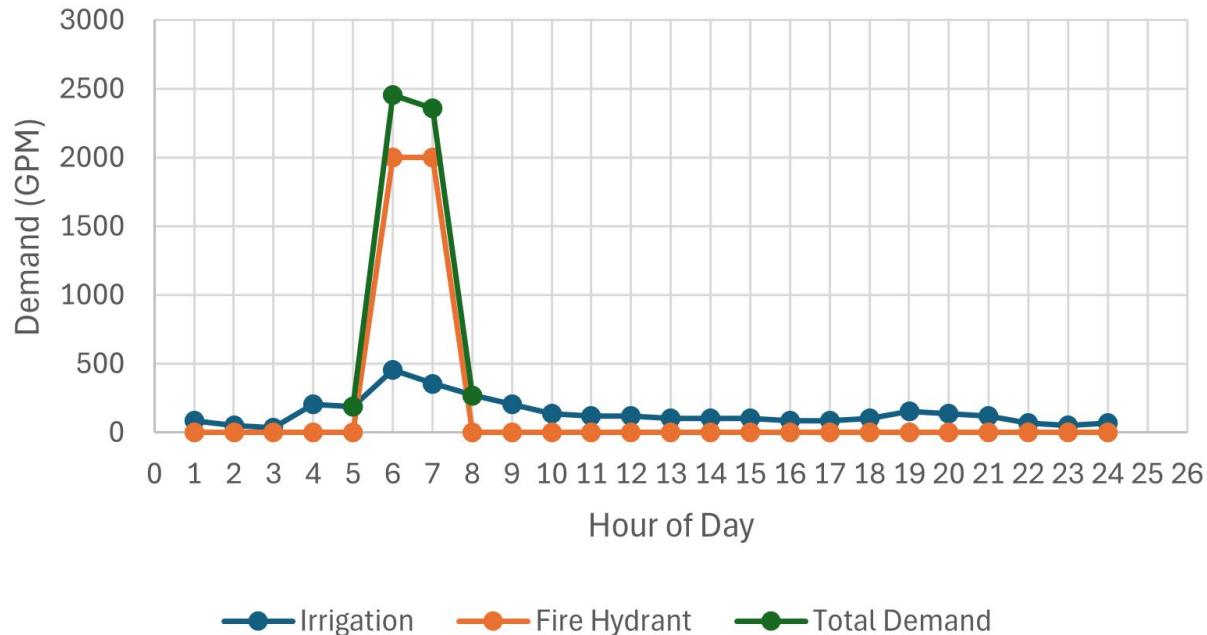


Hourly Pressure (For Fire Hydrant Node , 10 days)



Supply and Demand of System (Non-Potable)

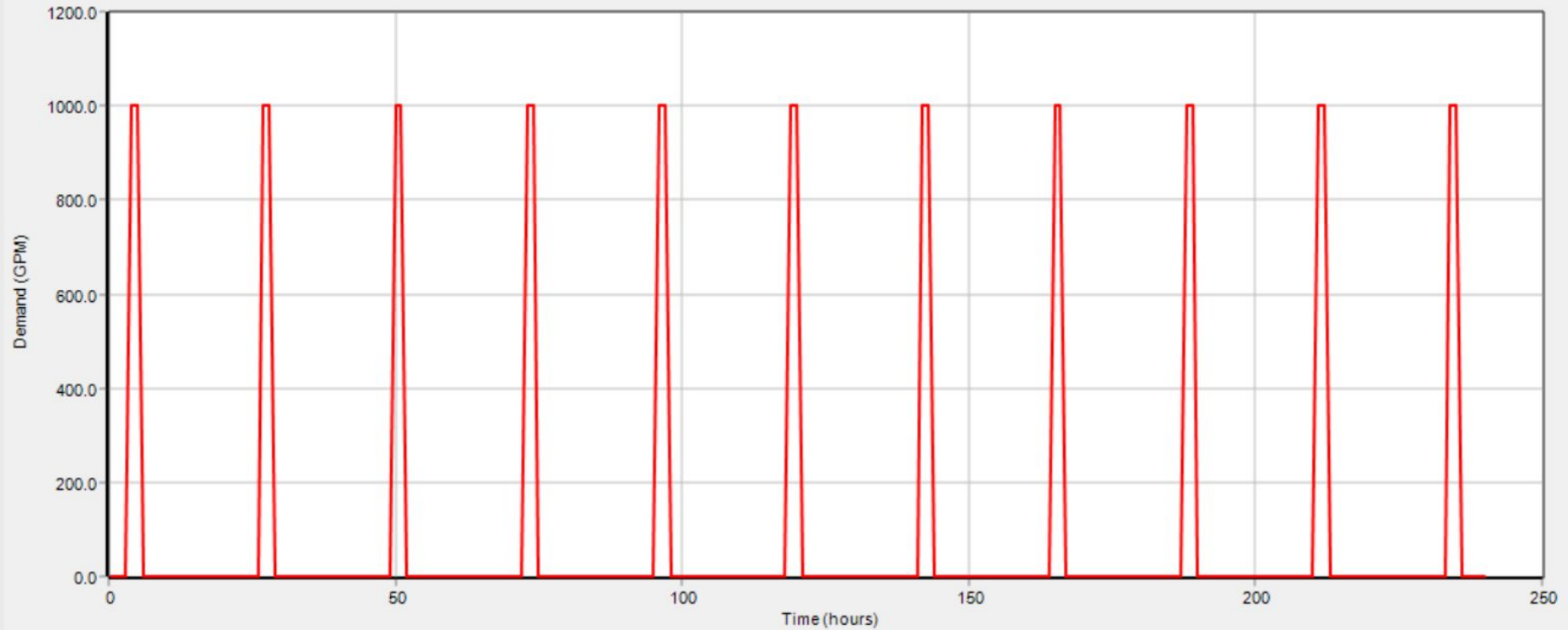
Total Non-Potable Demand vs Hour of Day



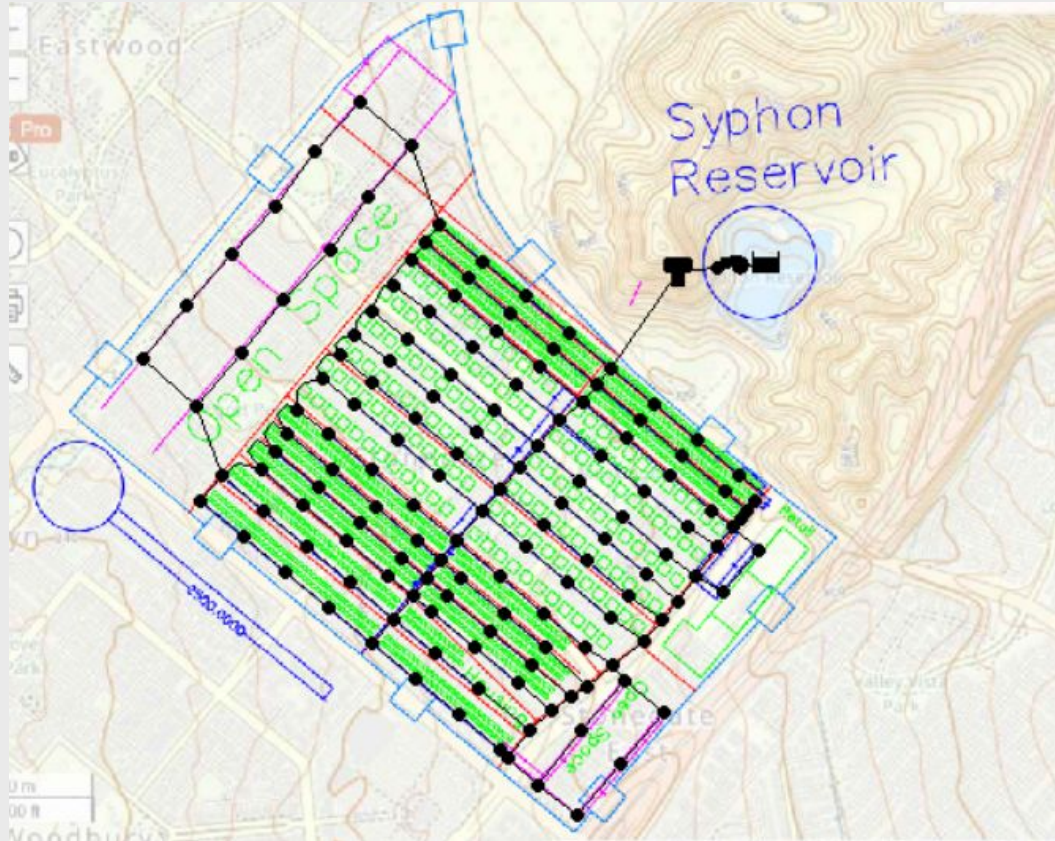
Details.

- 136 Non-potable Nodes, 124 as fire hydrant nodes, and 12 nodes for pure irrigation in open space section.
- Total Demand for the whole day reaches 5,400 gallons.

Fire Demand Node (10 Days)



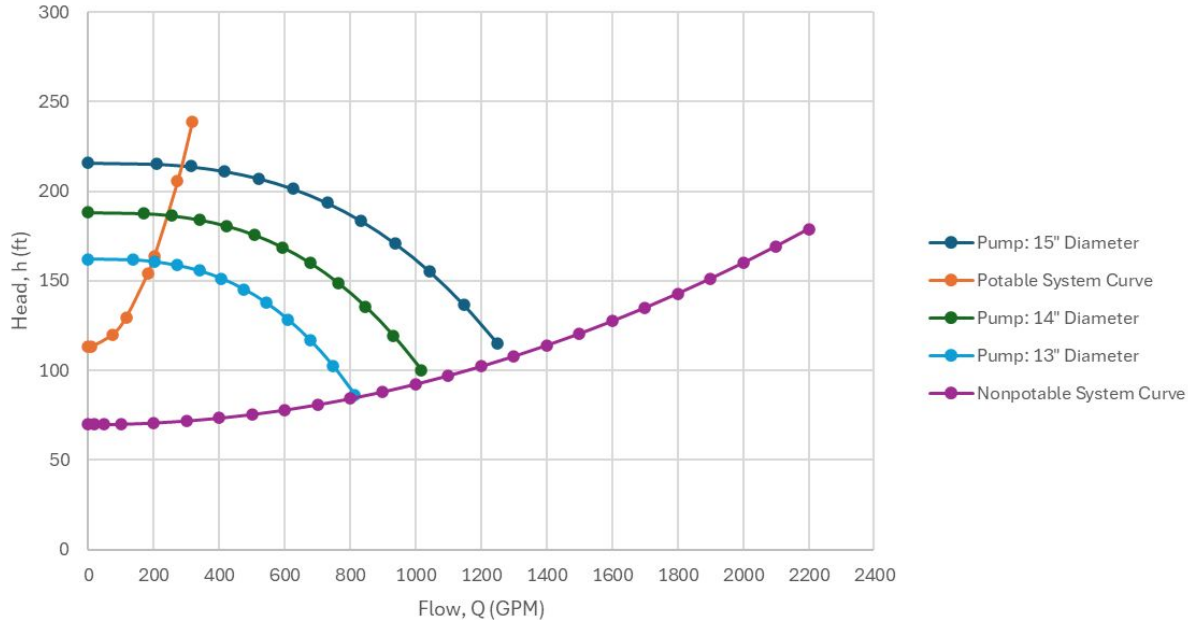
Design of the Non Potable Network



- Storage tank to store water for peak hours
- Open space in one area for easy irrigation
- No pipes under houses
- Pipe diameter coming into system = 7.5" to meet fire demand

Pump Design and Scaling

Flow vs Head: System and Pump Curve

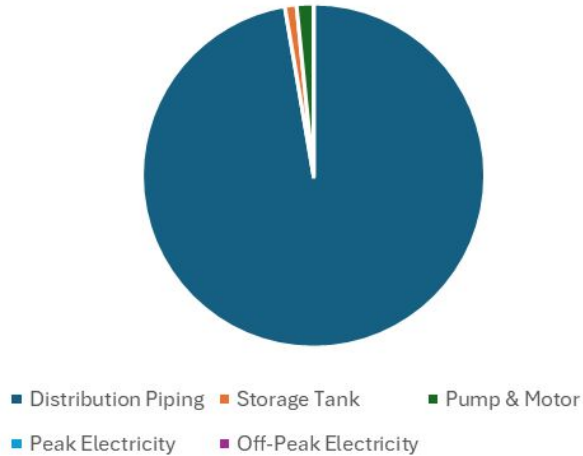


- Potable/Non-potable System Head Difference: 128ft & 70ft
- Sufficient Head: Pump Impeller $D \geq 13"$
- ω is constant 1760 rpm
- Several smaller pumps in series/parallel would not be cost-efficient
 - Current system contains single 14.25" impeller pump for the potable and 15.25" impeller for non-potable network

Cost Analysis of the Potable System



Potable System Cost Breakdown

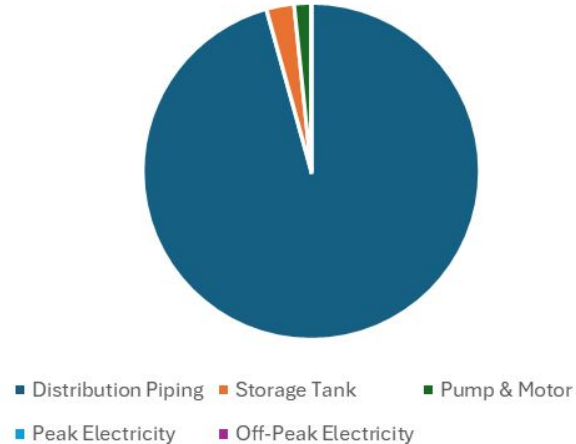


Distribution Piping	\$1,520,497
Storage Tank	\$17,276.09
Pump & Motor	\$24,875
Peak Electricity	\$23.21
Off-Peak Electricity	\$9.12
Total Capital Cost	\$1,562,648

Cost Analysis of the Non Potable System



Non Potable Network Cost Breakdown



Distribution Piping	\$1,540,790
Storage Tank	\$42,941.50
Pump & Motor	\$26,375
Peak Electricity	\$31.33
Off-Peak Electricity	\$12.30
Total Capital Cost	\$1,610,106

LID/Sustainability



- Minimal power usage within the pump system
- Minimal cut and fill utilization
- Pipe lengths were reduced as much as possible while still providing water to the project area
- Large open space for the community

Thank You!

